

DenseNet201-Powered Dental Radiograph Classification: A Deep Learning Approach for Multi-class Dental Pathology Detection

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Abstract--- In this study, the automated dental radiograph classification problem was addressed by a deep learning approach using DenseNet201 architecture. The research utilized a comprehensive dataset of 29,815 dental X-ray images across five categories: 69.0%, fillings (20.3%), implants (6.9%), cavity (2.1%), and impacted teeth (1.7%). To tackle class imbalance, the dataset was pre-processed with resizing to 224×224 pixels, normalization, and data augmentation techniques. 85.3% of the data were used for training, 9.1% for validation, and 5.6% for testing. With ImageNet weights and using Adam optimizer, the model achieved an overall accuracy of 91%. Performance analysis revealed excellent results for normal cases (F1-score: 0.95) and implants (F1-score: 0.91), while cavity detection showed lower performance (F1-score: 0.27) due to limited samples. While showing the capabilities of deep learning in dental radiography classification, the study also identifies where deep learning can improve on minority class detection. Finally, future work is proposed to incorporate attention mechanisms and increase the dataset to improve diagnostic accuracy in clinical applications.

Keywords: Dental Radiography, Deep Learning, DenseNet201, Medical Image Classification, Transfer Learning, Class Imbalance.

I. INTRODUCTION

The rapid development of deep learning techniques has contributed significantly to the field of medical imaging. Dental radiography is only a part of this. Panoramic radiographs and cone-beam computed tomography scans act as primary assistants in diagnosing dental diseases and performing oral health assessments. Certain dental conditions identified from these images always keep classification tasks challenging due to the inherent existence of several challenges like imbalance within a dataset. This has raised increasing interest in the application of deep learning algorithms for such tasks, particularly those related to multi-condition diagnosis, where one radiograph may point to multiple dental problems. Recent studies investigated the efficiency of deep learning methods in classification within dental conditions. Amongst them, a study conducted by Yilmaz et al. [1] is on the use of deep learning techniques in dental panoramic radiographs to classify teeth into different categories. Their work involved segmenting and identifying different types of teeth, thus demonstrating the capacity of CNN in handling complicated dental images. Minoo and Ghasemi [2] also designed some automated dental disease classification systems using deep learning models. They were able to classify dental cavity and gum diseases, among many others, using their methods,

therefore reducing the intricacies involved in diagnosis. According to Esmaeilyfard et al. [3] one of the most critical challenges in dental radiograph classification is the detection and classification of dental caries in CBCT images. In their study, deep learning techniques were used to deal with the difficulty of distinguishing between different stages of dental caries, further underscoring the need for high-quality, accurate models that can handle the subtleties of oral health conditions. Besides, Kokomoto et al. [4] proposed a new two-stage deep learning approach for the automatic dental age calculation from panoramic radiographs. This technique coupled object detection with image classification for the estimation of dental age, exemplifying how deep learning can be used to derive more complex diagnostic insights from a single image. Similarly, Ong et al. [5] proposed a completely automated approach for deep learning to staging dental development in panoramic radiographs. The approach described is now an efficient and reliable tool against assessing the developmental stage of the teeth with much higher consistencies compared to human judgment raters, which deep learning strengthened in its place for high automation and accuracy in diagnostics pertaining to dental studies. Although these works present promising progress with respect to the classification of dental radiographs, an imbalanced dataset-that one condition or set of conditions is usually underrepresented against the others-presents a challenge. Specifically, effective deep learning training for this multi-condition problem is critical; otherwise, it could easily lead to biased predictions, which need to determine multiple dental conditions at a time. This paper is devoted to the investigation of how deep learning algorithms, with their advanced architectures and techniques, will help to overcome these imbalances and improve multi-condition diagnosis accuracy and reliability on dental radiographs.

II. LITERATURE REVIEW

Deep learning approaches for dental diagnostics have recently gained considerable attention due to their potential for automatically and precisely detecting conditions from radiographs. Although the application of artificial intelligence is rapidly increasing to analyze images in dental informatics, such as panoramic radiographs, CBCT, and orthopantomograms, several challenges have to be overcome in the process, among which are issues such as dataset imbalance, multi-condition diagnosis, and the need for high accuracy in disease prediction. This work, therefore, reviews current improvements in deep learning over dental radiograph classification and its attempts to surmount the challenge of

imbalanced class in the diagnosis of multi conditions. Among them, one is the big contribution of the overall review comparison of deep learning architectures for dental diagnostics. The work by Hasnain et al. [6] presented the systematic review of different deep learning models for dental disease prediction via X-ray imaging. Their work compared a number of architectures while showing strengths and weaknesses of each model, especially for complicated dental conditions. They also emphasized the point that robust models should be developed that could handle the inherent challenges posed by dental radiograph data, such as imbalance and class overlap. Bouali et al. [7] go further in deep learning for dental diagnosis by discussing trends, applications, and challenges. They also indicated that the integration of deep learning and radiographic imaging has recently gained wide acceptance in various diagnostic fields like caries detection, periodontal disease diagnosis, and implant planning. They noted that deep learning models have shown much promise but also are faced with problems such as quality, small sample sizes, and class imbalance in order to improve the reliability of these models for multi-condition diagnosis. Another important direction of research is the elaboration of more specialized models for a particular condition. Thus, Ariji et al. [8] investigated how segmentations and detection models together increase the classification performance regarding the presence of cyst-like lesions on panoramic radiographs. Their findings indeed proved that the use of combined models increased the performance of classifying some lesions that would have been very difficult to tell apart from normal anatomic structures. This study therefore underlines the need to combine different deep learning methods in complex dental images for better classification. Meanwhile, Kurtulus et al. [9] designed a robust deep learning model for dental implant brand classification. This study pointed out that deep learning could be applied to carry out classification tasks other than disease detection, hence making AI applications on dental radiographs broader. Their research states that a well-trained model can also help in decision-making for specific dental appliances like implants, which might require discrimination among various brands or types-a task crucially important for personalized dental care. The technique utilizes Faster R-CNN by Laishram and Thongam [10] for the detection of teeth and classification from radiographs. The two use this deep learning-based technique to present an automated model that performs automatic detection and classification on orthopantomogram images. Therefore, it detects correct identification of structures of the teeth either due to the nature of overlaps or when the view from a radiograph is distorted. This is evidence of the effectiveness of this model, considering the increasing capability of deep learning in radiograph analysis, especially for cases that are difficult to be analyzed using traditional methods. In a somewhat related approach, Can et al. [11] proposed the CVAPool deep learning method, that makes use of the null-space of CNN weights to classify tooth diseases. Their novel approach showed the power of pooling techniques that could deal with highly imbalanced datasets, which is a common issue in dental image classification tasks. By focusing on the null space of CNN weights, their method effectively managed class imbalance, improving the accuracy of predictions for less frequent dental diseases. Other domains in which deep learning has had considerable promise include the segmentation of dental radiographs. Various approaches were reviewed by Brahma et al. [12] in a critical systematic review of the literature that presented the role of convolutional neural networks in dental radiography segmentation. Their

paper highlighted how different segmentation techniques may be used in the automatic detection of dental conditions while placing great emphasis on the application of deep learning principles to automate such tasks. While the features can be complex in dental images, CNNs have widely been used for the segmentation of regions of interest like cavities or lesions to enable more precise diagnosis with greater speed. Guler Ayyildiz et al. [13] compared several deep learning methods for the detection of different stages of periodontitis from dental radiographs in the field of periodontal disease detection. Their study showed that deep learning models could be trained to differentiate between mild, moderate, and severe stages of periodontitis, hence giving dentists an insight into the stages of the disease. This kind of automatic diagnosis could help in early detection and thus reduce the risk of further damage. For a series of case studies related to the automated identification of dental implant systems from highly degraded and noise-dominated radiographs, Lee et al. [14] presented an AI-centric approach that incorporates deep learning-trained models developed on a high-volume multicenter dataset in order for the model generalization against different conditions: imaging protocol, patient pathology variability, and different image quality ranges to handle conditions and variations resulting from the X-ray radiography process under both acquisition and object conditions. In this regard, this report strengthens the suggestion that large-sized and very diverse datasets assure the improvement of the strength of deep machine learning in order to validate appropriate performance after its clinical execution in hospitals, where an image captured may be blurry to noise. Yoo et al. [15] proposed the cascaded deep learning network called CACSNet, which automatically detects and classifies carotid artery calcification on panoramic radiographs. The performance of their model has already proved to be high and even robust when detecting small or not obvious calcifications, which might not have been detected by using classic diagnostic methods. This research highlights the increasing role played by deep learning in diagnosis under several conditions, especially where the models need not only to detect dental diseases but also systemic conditions related, thereby enhancing the overall diagnostic capability of radiographic imaging. Works reported in the literature on the classification of dental radiographs by deep learning bring out that AI has huge potential regarding the described challenges in the case of multi-condition diagnosis.

III. METHODOLOGY

A. Data Collection and Data Preprocessing

The dataset used in this research was obtained from a public repository on Kaggle [16], a repository that contains a rich collection of dental radiographs for classification tasks. It consists of a total of 29,815 images distributed across five categories: The dataset consists of normal (20,582 images), fillings (6,047 images), cavity (641 images), implants (2,047 images), and impacted teeth (498 images), as seen in Table I.

TABLE I. Distribution of different tooth conditions

Category	Number of Images	Percentage
Normal	20,582	69.0%
Fillings	6,047	20.3%
Implants	2,047	6.9%
Cavity	641	2.1%
Impacted Teeth	498	1.7%
Total	29,815	100%

This dataset provides diverse dental conditions to support the development and evaluation of classification models. The dataset images underwent several preprocessing steps to ensure optimal model performance. Images were resized to 224×224 pixels and normalized to [0,1] range. Data augmentation techniques including random horizontal flipping, rotation ($\pm 15^\circ$), and brightness adjustment ($\pm 10\%$) were applied to enhance model generalization. Class weights were implemented to address the significant class imbalance in the dataset, as shown in Fig 1.

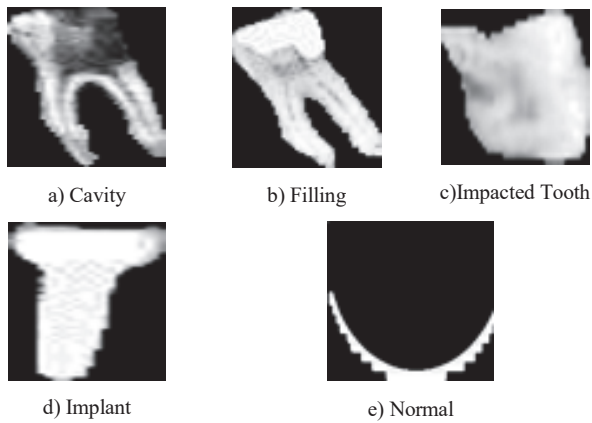


Fig. 1. Dataset Images for a) Cavity, b) Filling, c) Impacted Tooth, d) Implant, e) Normal

B. Data Splitting

A dataset of 29,815 dental radiographic images was split into training, validation and testing sets in a strategic way to guarantee robust model evaluation. The training set consists of 25,430 images (85.3% of the total dataset), having 17,400 normal, 5,242 filling, 1,784 implant, 576 cavity, and 428 impacted teeth images. It contains 2,721 images (9.1%), distributed with 1,941 normal, 540 filling, 159 implant, 43 cavity, and 38 impacted teeth cases in the validation set. The testing set contains 1,714 (5.6%) images including 1,241 normal, 315 filling, 104 implant, 22 cavity and 32 impacted teeth radiographs. This split keeps the approximate class distribution ratios across all three sets, so it samples representative but has enough data to train and evaluate the model, as seen in Table II.

TABLE II. Distribution of Training, Testing and Validation data

Category	Training	Validation	Testing
Cavity	576	43	22
Normal	17,400	1,941	1,241
Fillings	5,242	540	315
Impacted Teeth	428	38	32
Implants	1,784	159	104

C. Model Selection and Architecture

In this study, the DenseNet201 architecture is used, a popular, powerful pre-trained model used for image classification owing to its dense connectivity that improves gradient flow and reduces parameter redundancy. The model from ImageNet weights are used as initialized weights, except for the top classification layer, to tailor it to the dental radiography dataset. Feature extraction is made with DenseNet201 layers, followed by a Global Average Pooling layer to create spatial dimensions to a minimum and fully connected layers with Dropout to prevent overfitting. The last

layer uses softmax activation to determine what class image belongs to out of 5. Class imbalance is addressed during training by incorporating class weights. The resulting model design guarantees effective transfer learning, optimizes the classification of dental radiographs, and guarantees robust performance on the challenging dataset, as seen in Table III.

TABLE III. Model Summary for the Architecture

Layer Type	Output Shape	Number of Parameters
Input Layer	(None, 64, 64, 3)	0
DenseNet201 (Pretrained)	(None, 64, 64, 201)	20,045,824
GlobalAveragePooling2D	(None, 201)	0
Dense (256 units)	(None, 256)	51,776
Dropout (0.3)	(None, 256)	0
Dense (128 units)	(None, 128)	32,896
Dropout (0.2)	(None, 128)	0
Dense (32 units)	(None, 32)	4,128
Dense (5 classes)	(None, 5)	165
Total Parameters		20,134,889

D. Training Process and Evaluation

To prevent overfitting, the model was trained with the Adam optimizer and a learning rate scheduler and early stopping. Having multi-class classification, the loss function used was sparse_categorical_crossentropy, and the metric was accuracy. To address dataset imbalance, the dataset was trained using class weights. The training took place for maximum of 100 epochs with batch size of 256. Training halted early with early stopping where if the validation loss did not improve for 20 consecutive epochs, then restoring the best weights. During validation loss plateau was noticed, learning rate was reduced by factor 0.0.1 using learning rate reduction callback. Furthermore, ModelCheckpoint was used to save the best model for later evaluation. The confusion matrix, classification report and F1 score are used to evaluate the performance of the model and to evaluate how well is the model classified across all the classes present in the dataset. This is important in medical imaging where high precision and recall are prime importance. With transfer learning, class weighting, and advanced regularization techniques, the model is expected to perform well in classifying dental radiography data even with an imbalanced dataset using the DenseNet201 architecture.

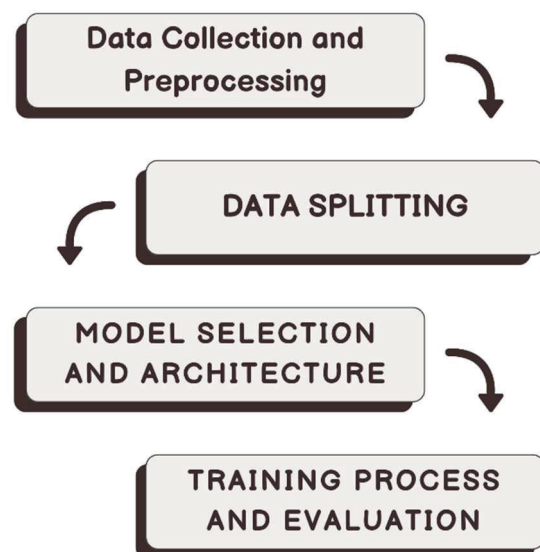


Fig. 2. Proposed Methodology used in this study

IV. RESULTS

A. Training and validation loss

Loss curves provide insight into the model's learning dynamics. The training loss decreases from 1.3038 to 0.5280 in the first two epochs and decreases more and more quickly but finally achieving a minimum at around 0.0425 (epoch 43). This shows that the training set is being optimised properly. However, the range of validation loss is much larger (higher) than the range of training loss. These accuracy dips are marked by notable spikes which occur at epochs 4 (1.1530) and 35 (1.2746). Further, the fact that after epoch 20 there is still no closer gap between training and validation loss indicates that overfitting is likely a concern. The erratic behavior in validation loss suggests the model isn't consistent with generalization. At epoch 25, the validation loss is at the lowest level of 0.3027, which is likely a good stopping criterion compared to later epochs. This pattern indicates that the model improves further on the training data but may actually be learning less generalizable features in the later epochs, as shown in Fig 3.

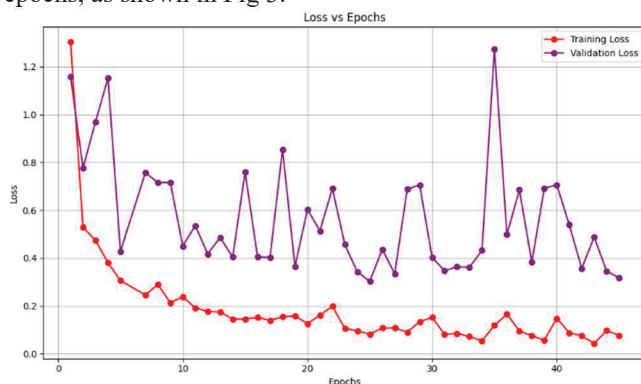


Fig. 3. Training and validation loss

B. Training and Validation Accuracy

The training accuracy is steadily climbing up, from its initial 38.28% and getting to impressive 97.24% around epoch 43. It is an indication of the model learning on the training data by effectively. But the validation accuracy is more volatile, ranging from 60.38% to 92.10%. Epochs 4 (60.38%) and 35 (66.70%) have notable dips indicating that the model temporarily lost generalization capability. Later epochs, training and validation accuracy grow apart with training accuracy always above 90% but validation accuracy around 85-90%. This is a classic overfitting signature; the model fits the training data better than unseen validation data. Validation accuracy reaches its peak of 92.10% at epoch 42 which may be a reasonable point for model selection. The oscillating accuracy pattern also points into the direction of possible use of techniques such as learning rate scheduling and regularization to stabilize the performance of the model, as shown in Fig 4.

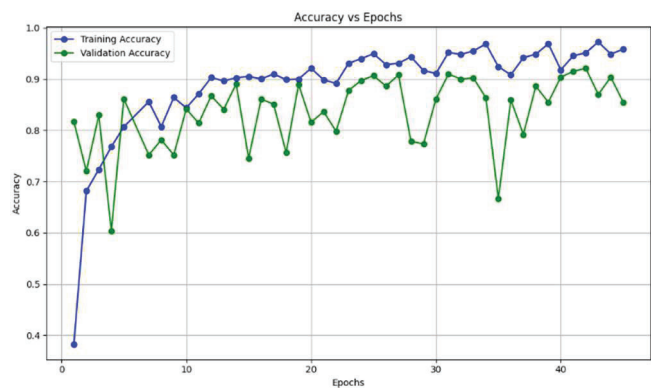


Fig. 4. Training and validation accuracy.

C. Confusion Matrix Analysis

The provided confusion matrix shows the model's performance across five categories: Cavity, Fillings, Impacted Tooth, Implant, and Normal. The model is well-performing in the "Normal" class, correctly identifying 1069 out of 1096 instances, but in other categories, it would fail. The model misclassifies between "Cavity" and "Fillings" and often confuses "Cavity" and "Normal" as well. Confusion exists in both the "Implant" class, as it is misclassified as "Cavity," "Fillings," and "Normal." Furthermore, there is a general problem in determining whether a condition is an "impacted tooth," a "cavity," or "fillings." These issues indicate that the model could be further refined: better feature engineering, more training data diversity, or more complicated model architecture may be necessary to increase accuracy across all categories, as illustrated in Fig 5.

		Confusion Matrix				
Actual	Cavity	8	15	2	1	14
	Fillings	4	252	3	2	25
	Impacted Tooth	2	3	19	1	3
	Implant	1	2	2	80	3
	Normal	4	15	3	3	1069
		Predicted				
		Cavity	Fillings	Impacted Tooth	Implant	Normal

Fig. 5. Confusion matrix showing classification performance.

D. Classification Report Analysis

The classification report for the dental radiography model reflects a good performance across five classes: Fillings, Normal, Impacted Tooth, Cavity, and Implant. The model reaches to 91 % in overall accuracy, signifying a very high level of precision in the predicted. Interestingly, the 'Normal' class has the highest precision and recall, with 0.96 and 0.95, respectively, which is expected as it has the highest support and better distinction in the dataset. Strong performance by the classes of the "Fillings" and "Implant" can also be seen, with precision and recall values around 0.88 to 0.92, that demonstrates the model is also working these two classes. However, lower precision and recall scores for the "Cavity" and "Impacted Tooth" classes demonstrate it poses challenges. That may also be due to the fact that they have a smaller support and more complex features that overlap other classes. While there is obvious challenges to the model, the

macro average is balanced statewide across all metrics, indicating the model has no reason to show bias to class distribution. Overall, this classification report shows the model's robustness for dental radiography classification, with reliable diagnostic support and areas of further refinement, especially for less frequent or more complex categories, as seen in Table IV.

TABLE IV. Classification Report

Class	Precision	Recall	F1-Score	Support
Cavity	0.20	0.42	0.27	19
Fillings	0.88	0.88	0.88	287
Impacted Tooth	0.80	0.65	0.72	29
Implant	0.90	0.92	0.91	87
Normal	0.96	0.95	0.95	1114
Accuracy			0.91	2560
Macro Avg	0.75	0.76	0.75	2560
Weighted Avg	0.92	0.91	0.92	2560

V. CONCLUSION

In this paper, a strong deep learning model for dental radiography classification is proposed using DenseNet201, which achieved a great accuracy of 91%. Performance on normal cases and dental fillings is excellent while the precision and recall for implant detection is high. By assessing the model across five distinct classes (Cavity, Fillings, Impacted Tooth, Implant, and Normal) the model can perform well across a range of complex medical imaging tasks. In particular, the model performs very well in classifying Normal (95% F1 score) and Implant (91% F1 score) cases but struggles to identify Cavity cases (27% F1 score) since there are very few samples and the feature space is complex. The training progression exhibits continuous improvement which peaks at epoch 42 and shows little overfitting concern from the validation metrics. In future work, attention mechanisms and multi-modal approaches could be incorporated to improve diagnostic accuracy. In addition, more complex architectures and an expanded dataset may also improve the model's performance in clinical applications. A model's utility in real world settings will be validated through conducting prospective clinical trials.

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